



Loop Engineering

*Stop prompting your agent every time.
Design the system that prompts itself.*

 Heartbeat

 Cron

 Hook

 Goal

Swipe to learn all 4 types + when to use each →

by Sweta · Agentic Data Lab



What Is Loop Engineering?

**"You don't prompt the agent.
You design the system that prompts the agent."**

— Addy Osmani, June 2026



Without Loop Engineering

1. You write every prompt manually
2. You wait, review, re-prompt
3. You are the revision cycle
4. You can't step away



With Loop Engineering

1. You define goal + stop rule
2. Agent acts autonomously
3. Agent self-verifies output
4. Loop repeats until done

Loop engineering = the shift from prompt engineering to system engineering

@agenticdatalab



4 Types of Agent Loops

Every production AI agent uses one or a combination of these



Heartbeat

"Always-on watcher"

Best for:

Monitoring, queues, status checks

→ *Deep dive on next slides*



Cron

"Reliable scheduler"

Best for:

Reports, digests, nightly tasks

→ *Deep dive on next slides*



Hook

"Reactive listener"

Best for:

Webhooks, events, user actions

→ *Deep dive on next slides*



Goal

"Autonomous achiever"

Best for:

Research, debugging, open tasks

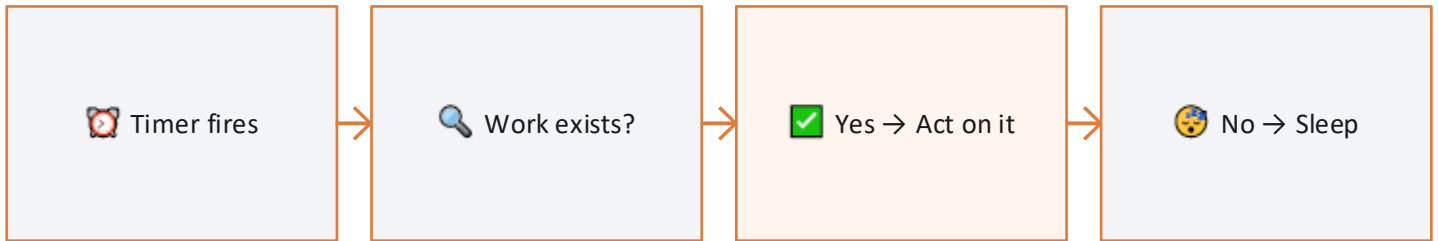
→ *Deep dive on next slides*



Heartbeat Loop

"The always-on watcher"

The agent wakes on a fixed interval — every 5 min, every hour — checks if there's work, acts on it, then sleeps. It's not always running, just always available.



↺ repeats on every tick



Use Heartbeat When...

- Monitoring a support queue
- Watching a live data stream
- Checking GitHub PRs on schedule
- Triage + escalation systems



Avoid When...

- Task needs deep reasoning
- Quality matters more than speed
- Budget is tight (high fixed cost)
- One-shot tasks with no repeat



Heartbeat is the cheapest loop — cost bounded by how often it ticks, not how long it runs



Cron Loop

"The reliable scheduler"

Fires at a precise calendar time — daily 9 AM, every Monday, first of the month. Unlike heartbeat, it doesn't poll constantly. It fires once on schedule and completes a full task run.

Heartbeat vs Cron — the key difference

Heartbeat

Polls every N minutes, checks if work exists first

Cron

Fires at exact time, always runs the full task

Use Cron When...

- Weekly analytics report
- Daily LinkedIn post scheduling
- Monthly client billing summary
- Nightly data pipeline runs

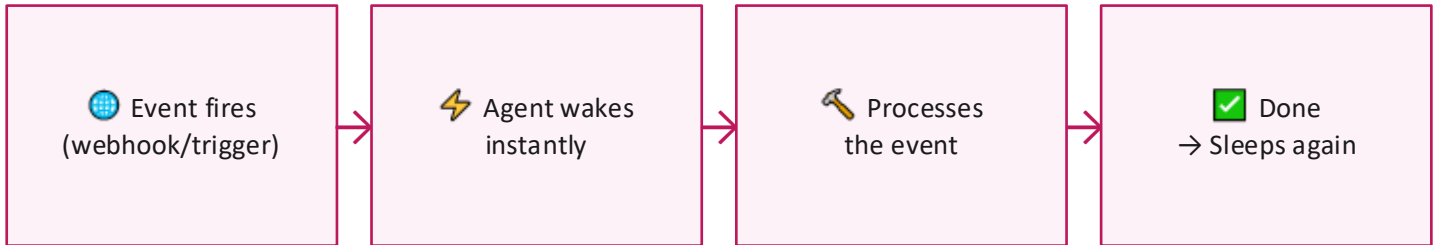
Avoid When...

- Real-time response needed
- Event is unpredictable
- Task duration is unknown
- Frequency needed $< 1/\text{day}$

? Hook Loop

"The reactive event listener"

No polling. No schedule. Agent sleeps completely until an external event fires — webhook, commit, message, form — then wakes, acts immediately, then sleeps again. Zero idle cost.



Real-World Hook Trigger Examples

 New email → AI triage + label

 Git push → AI code review

 New Upwork job → AI proposal

 Order placed → AI fulfilment note

Use Hook When...

- Real-time response critical
- Events are unpredictable

Avoid When...

- No external event source
- Batch processing preferred



Goal Loop

"The autonomous achiever"

No schedule, no trigger. You define a goal + stop condition. The agent iterates — act → verify → decide → repeat — until its success check returns true or a budget ceiling hits.

⚠ **The #1 Mistake: Vague Goals = Infinite Loop + Empty API Budget**

❌ Bad Goal

"Research AI agents and write something good"

No stop condition → loops forever → burns budget

✅ Good Goal

"Research 5 competitors, extract 3 gaps each, return structured JSON"

Clear, binary, testable → agent knows exactly when to stop



Use Goal Loop When...

- Research + summarise tasks
- Autonomous debugging / fixes



Avoid When...

- Goal can't be defined clearly
- Token/budget ceiling unknown



Which Loop Should You Use?

Quick decision framework — answer the question, pick the loop

Q: Need real-time response to an event?

→ Hook Loop

Q: Task repeats on a fixed calendar schedule?

→ Cron Loop

Q: Need to continuously poll + check status?

→ Heartbeat Loop

Q: Goal defined but path to get there unknown?

→ Goal Loop

Q: Multiple behaviours running simultaneously?

→ Combine 2–3 loops

💡 Most production agents combine patterns — e.g. Hook + Heartbeat + daily Cron report

@agenticalab



Research Agent: Dual Loop in Action

Implemented with LangGraph · Claude Haiku + Sonnet · Real code walkthrough

Research Agent — Heartbeat + Goal Architecture

❤️ Heartbeat Loop

- 🕒 Wakes every 30 min
- 🔍 Checks for new topics
- 📁 Queues topic if found
- 😴 Sleeps till next tick



🎯 Goal Loop

- 📋 Goal: Research topic deeply
- 🔍 Search → Read → Summarise
- ✅ Stop: 5 quality sources
- 📄 Returns structured brief

Heartbeat detects work → hands off to Goal Loop → Goal completes research → returns to Heartbeat

Stack used to build this

⚙️ LangGraph

⚙️ Claude Haiku

⚙️ Claude Sonnet

⚙️ DuckDuckGo Search

⚙️ FastAPI

⚙️ SQLite State



Full implementation walkthrough — link in comments ↓



Loop Engineering is the biggest shift in AI agents since prompt engineering.

I built Heartbeat + Goal loops live in LangGraph
with Claude Haiku + Sonnet. Real code. Real failures.



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New video every Wednesday & Saturday



Save this ·  Repost to a dev who still hand-prompts their agents

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